

# Sequence Models II: Encoder–Decoder, Teacher Forcing and Attention

Mapping one sequence to another — and letting the decoder look back

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ES 667 · Deep Learning · IIT Gandhinagar

# **From one sequence to another**

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# Where we are

The last three lectures modeled **one** sequence — next-token prediction over  $x_{1:T}$ :

$$p(x_{1:T}) = \prod_{t=1}^T p(x_t \mid x_{<t})$$

This lecture changes the **task**: map an **input** sequence to a **different output** sequence — translation, summarization, speech. Input and output can even have **different lengths**.

# Tasks that map sequence $\rightarrow$ sequence

Many problems are  $x_{1:T} \rightarrow y_{1:U}$  — a source sequence in, a target sequence out:

| <b>Task</b>        | <b>Input <math>x_{1:T}</math></b> | <b>Output <math>y_{1:U}</math></b> |
|--------------------|-----------------------------------|------------------------------------|
| translation        | English sentence                  | French sentence                    |
| summarization      | a document                        | a short summary                    |
| speech recognition | audio frames                      | a transcript                       |
| image captioning   | image regions                     | a caption                          |
| dialogue           | conversation so far               | the next reply                     |
| forecasting        | past observations                 | future horizon                     |

one general shape — **sequence in, sequence out** — behind all of them

The defining difficulty: the output length  $U$  is **not** the input length  $T$ , and is **not known in advance**.

“I like cats” ( $T = 3$ )  $\rightarrow$  “j’aime les chats” ( $U = 3$ )

A token classifier ( $x_{1:T} \rightarrow y_{1:T}$ , one label per input) **cannot** do this: it forces  $U = T$  and a fixed alignment. We need a model that **generates** an output of **flexible** length.

# The conditional sequence model

Keep the autoregressive factorization — but now **condition on the whole source**  $x_{1:T}$ :

$$p(y_{1:U} \mid x_{1:T}) = \prod_{u=1}^U p(y_u \mid y_{<u}, x_{1:T})$$

**autoregressive** in the target: each  $y_u$  depends on the tokens generated **before** it.

**conditioned** on the source: every factor also sees the **entire input**  $x_{1:T}$ .

same next-token idea as Lecture 11 — now with a **source** to condition on

# The encoder–decoder

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# Two networks, one bridge

Split the job in two (Sutskever, Vinyals & Le 2014; Cho et al. 2014):

**Encoder** — reads the source  $x_{1:T}$  and compresses it into a **context vector**  $c$ .

**Decoder** — a conditional language model that **generates**  $y_{1:U}$  from  $c$ .

the context  $c$  is the only bridge — everything the decoder knows about the source is in  $c$

The encoder is an RNN (LSTM/GRU) run over the source; its **final state** is the context:

$$h_t = f_{\text{enc}}(h_{t-1}, e(x_t)), \quad t = 1, \dots, T$$

$$c = h_T \quad (\text{the whole source, summarized in one vector})$$

read the source left to right;  $h_T$  is a fixed-size summary of **all** of it

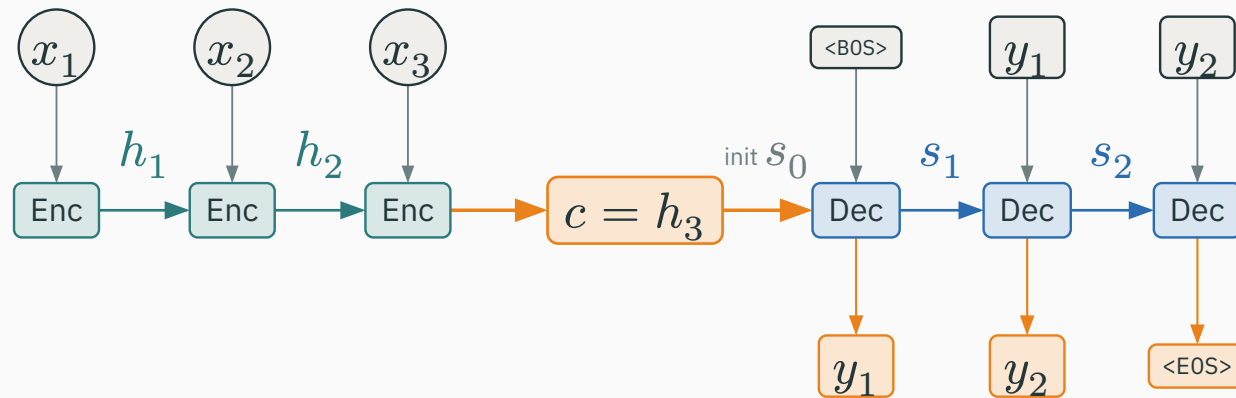
A second RNN, **initialized from  $c$** , generates the target one token at a time:

$$s_u = f_{\text{dec}}(s_{u-1}, e(y_{u-1}), c), \quad s_0 = c$$

$$p(y_u \mid y_{<u}, x_{1:T}) = \text{softmax}(W_o s_u + b_o)$$

the decoder is a language model over the target, **conditioned** on the source through  $c$

Encode the source into  $c$ ; hand  $c$  to the decoder; generate the target autoregressively:



$\langle \text{BOS} \rangle$  starts generation; each output feeds the next step;  $\langle \text{EOS} \rangle$  stops it

Two special tokens make **variable-length** generation work:

<BOS> — the decoder's **first input**, so it can produce  $y_1$  with no target history.

<EOS> — an ordinary vocabulary token; **emitting it** is how the model **chooses to stop**.

Training a pair shifts the target by one

**input** vs **target** of the decoder:

|                |       |     |     |       |
|----------------|-------|-----|-----|-------|
| decoder input  | <BOS> | y_1 | y_2 | y_3   |
| decoder target | y_1   | y_2 | y_3 | <EOS> |

Maximize the conditional log-likelihood — sum token cross-entropy over the target:

$$\mathcal{L} = - \sum_{u=1}^U \log p(y_u \mid y_{<u}, x_{1:T})$$

Batches mix target lengths, so pad and **mask** the padded positions out:

$$\mathcal{L} = \frac{\sum_{i,u} m_{iu} (-\log p_{iu})}{\sum_{i,u} m_{iu}}, \quad m_{iu} \in \{0, 1\}$$

$m_{iu} = 0$  on <PAD> — padding is invisible to the gradient

# Teacher forcing

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At each decoder step the input is the **previous target token**. Which one — the **true** one, or the model's **guess**?

**Teacher forcing:** during training, feed the **ground-truth**  $y_{u-1}^{\text{true}}$ , not the prediction  $\hat{y}_{u-1}$ .

$$s_u = f_{\text{dec}}(s_{u-1}, e(y_{u-1}^{\text{true}}), c) \quad (\text{training})$$

**Stable, parallel supervision** — every step gets a **correct** input, so all  $U$  targets can be scored in one pass.

**Without it** — an early wrong token becomes the **next** step's input, and the error **compounds** down the sequence.

teacher forcing decouples the steps: each is trained as a clean next-token classification

The catch — the two regimes feed **different** previous tokens:

**train:**  $s_u = f_{\text{dec}}(s_{u-1}, e(y_{u-1}^{\text{true}}), c)$

**infer:**  $s_u = f_{\text{dec}}(s_{u-1}, e(\hat{y}_{u-1}), c)$

**Exposure bias:** the model only ever saw **true** histories in training, but at inference it must consume its **own** (sometimes wrong) outputs — a distribution it was never trained on.

A middle ground (Bengio et al. 2015): **sometimes** feed the model's own prediction during training.

$$\text{decoder input at step } u = \begin{cases} y_{u-1}^{\text{true}} & \text{with prob. } p \\ \hat{y}_{u-1} & \text{with prob. } 1 - p \end{cases}$$

anneal  $p : 1 \rightarrow 0$  over training — start fully teacher-forced, gradually expose the model to its own outputs

## **A worked seq2seq pass**

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# The toy translation

Translate “I like cats”  $\rightarrow$  “j’aime les chats”. Set up the source and the shifted target:

$x = [\text{I}, \text{like}, \text{cats}, \langle \text{EOS} \rangle]$

decoder input =  $[\langle \text{BOS} \rangle, \text{j 'aime}, \text{les}, \text{chats}]$

decoder target =  $[\text{j 'aime}, \text{les}, \text{chats}, \langle \text{EOS} \rangle]$

the target is the input shifted by one — teacher forcing supplies the true tokens

Run the encoder over the four source tokens; take the final state as the context:

$$h_1, h_2, h_3, h_4 = f_{\text{enc}}(\dots), \quad c = h_4$$

Initialize the decoder and take its **first** step from <BOS>:

$$s_1 = f_{\text{dec}}(s_0, e(\text{<BOS>}), c), \quad s_0 = c$$

$$p_1 = \text{softmax}(W_o s_1 + b_o) \quad (\text{a distribution over the French vocabulary})$$

The target for step 1 is `j'aime`; the loss is its negative log-probability:

$$\mathcal{L}_1 = -\log p_1(\text{j'aime})$$

Teacher forcing now feeds the **true** `j'aime` (not  $\hat{y}_1$ ) as the input to step 2:

$$s_2 = f_{\text{dec}}(s_1, e(\text{j'aime}), c)$$

each step: predict, score against the true next token, then feed that true token onward

Sum the per-step cross-entropies over the whole target:

$$\mathcal{L} = -\log p_1(\text{j' aime}) - \log p_2(\text{les}) - \log p_3(\text{chats}) - \log p_4(\text{<EOS>})$$

$$\mathcal{L} = -\sum_{u=1}^4 \log p_{u(y_u^{\text{true}})} \text{ — four next-token classifications, one per target position}$$

note the last term: the model is **supervised to emit <EOS>** — that is how it learns to stop

# **Inference and decoding**

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At inference there is no true history — feed the model's own output back. Simplest rule: take the **arg max** each step:

$$\hat{y}_u = \arg \max_v p(v \mid \hat{y}_{<u}, x_{1:T})$$

**Greedy is myopic.** The locally most-probable token need not start the globally most-probable **sequence** — one early commitment can doom the rest.

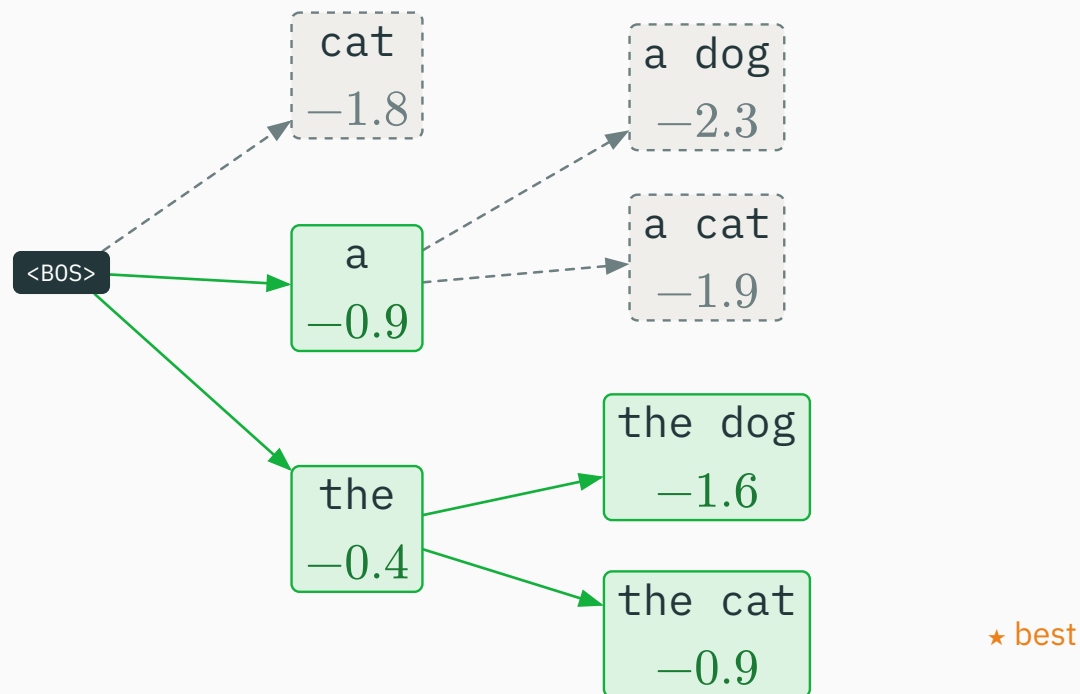
Keep the  $B$  best **partial** sequences (the **beam**) instead of one. Score a hypothesis by its log-probability:

$$\text{score}(y_{1:u}) = \sum_{j=1}^u \log p(y_j \mid y_{<j}, x_{1:T})$$

- **expand** every beam by all vocabulary tokens;
- **score** each extension; **keep** only the top  $B$ ;
- repeat until enough beams emit <EOS> or a maximum length is reached.

$B = 1$  is greedy; larger  $B$  searches wider (more compute) for a better sequence

$B = 2$ : at each level, expand the survivors, score, and keep the best two:



green = kept in the beam; dashed = pruned; the highest-scoring complete sequence wins

Every factor has  $\log p \leq 0$ , so a **longer** sequence has a **more negative** raw score:

$$\text{score}(y_{1:U}) = \sum_{j=1}^U \log p(y_j \mid \dots) \leq 0 \quad (\text{shrinks with each token})$$

Raw log-prob **favors short outputs** — the beam is biased toward emitting <EOS> early. Fix it by **length-normalizing** the score:

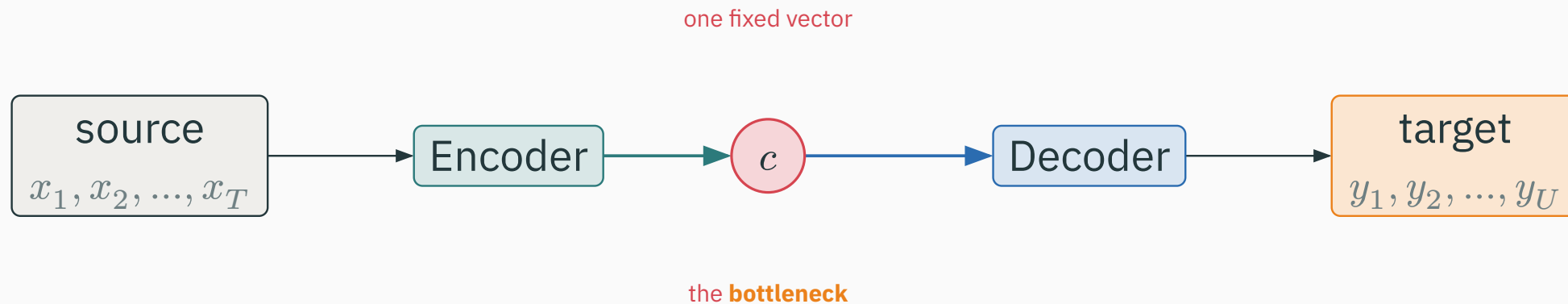
$$\text{score}_{\text{norm}}(y_{1:U}) = \frac{1}{U^\alpha} \sum_{j=1}^U \log p(y_j \mid \dots), \quad \alpha \in [0.6, 1]$$

# The fixed-vector bottleneck

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# Everything through one vector

In the plain encoder–decoder, **all** of the source must pass through a **single**  $c = h_T$ :



a 40-word sentence and a 4-word sentence get the **same** fixed-size  $c$

- for a **long** source, one vector cannot retain every detail — early tokens get **overwritten**;
- the decoder needs **different** parts of the source at **different** output steps — but  $c$  is **fixed** for all of them;
- empirically, translation quality **falls off sharply** as the source grows (Bahdanau et al. 2015).

The bottleneck is **architectural**, not a training issue: a single fixed vector is simply too small a channel for a long, structured source.

At **each** output step, the decoder should look at the **relevant** source positions — not a frozen summary:

**replace one fixed  $c$  with a per-step, source-dependent context  $c_u$**

when translating chats, look at cats; when translating j'aime, look at like — this is **attention**

# Attention



Keep **all** encoder states  $h_1, \dots, h_T$ . Build the context as a **weighted average** of them, recomputed every step:

$$c_u = \sum_{t=1}^T \alpha_{ut} h_t, \quad \alpha_{ut} \geq 0, \quad \sum_{t=1}^T \alpha_{ut} = 1$$

$\alpha_{ut}$  = how much output step  $u$  **attends to** source position  $t$  — a soft, differentiable selection

The weights come from a **score** between the decoder query and each encoder state, passed through a softmax:

$$e_{ut} = \text{score}(s_{u-1}, h_t) \quad (\text{one scalar per source position})$$

$$\alpha_{ut} = \frac{\exp(e_{ut})}{\sum_{j=1}^T \exp(e_{uj})} \quad (\text{softmax over source positions})$$

high score  $\rightarrow$  high weight  $\rightarrow$  that source state dominates the context

Three standard ways to score a query  $s$  against a key  $h$ :

| Name               | $\text{score}(s, h)$          | Note                                           |
|--------------------|-------------------------------|------------------------------------------------|
| dot                | $s^\top h$                    | no parameters; needs $\dim s = \dim h$ (Luong) |
| general / bilinear | $s^\top W h$                  | a learned interaction matrix $W$ (Luong)       |
| additive / MLP     | $v^\top \tanh(W_s s + W_h h)$ | a small MLP (Bahdanau)                         |

Bahdanau 2015 = **additive**; Luong 2015 = **dot** / **general** — all learned end-to-end

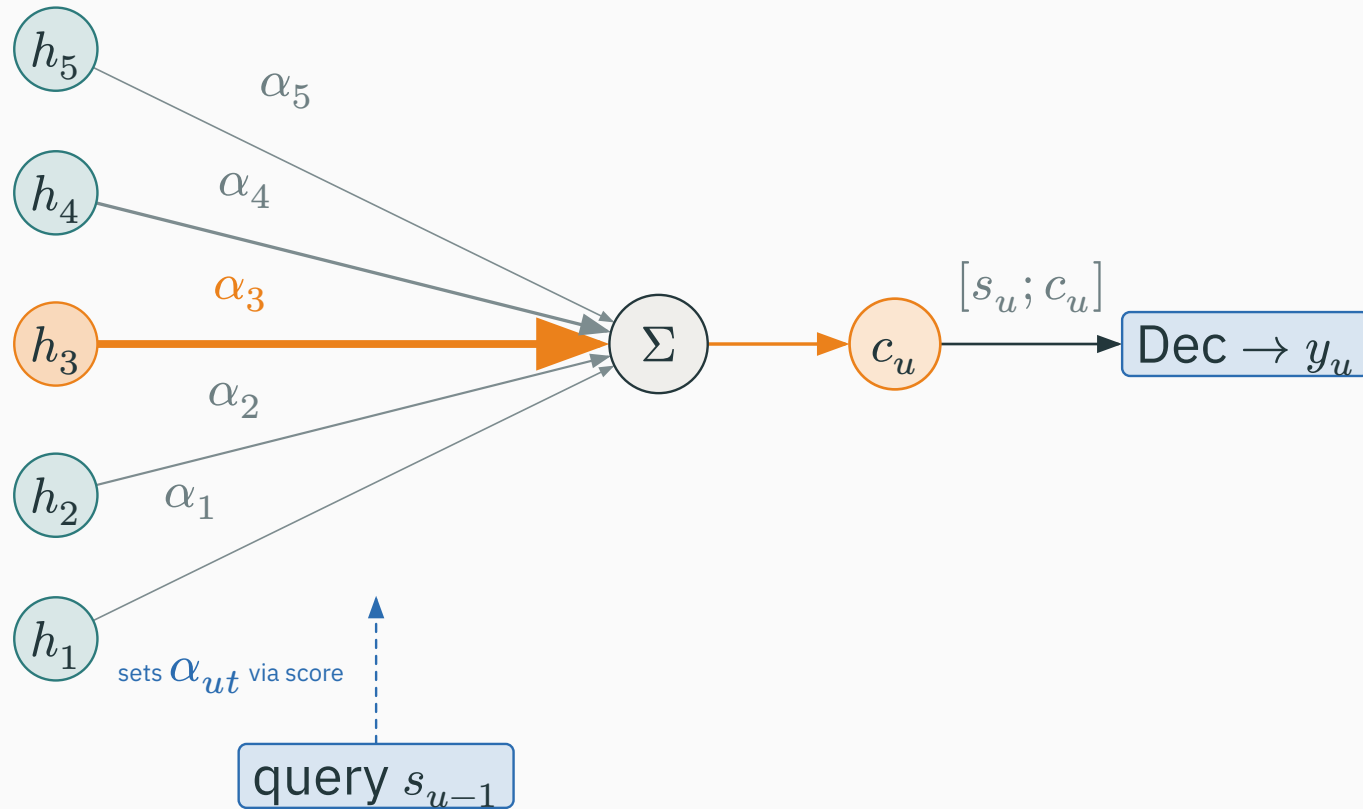
Combine the context with the decoder state, then read out the next token:

$$c_u = \sum_{t=1}^T \alpha_{ut} h_t$$

$$p(y_u \mid y_{<u}, x_{1:T}) = \text{softmax}(W_o [s_u; c_u] + b_o)$$

the readout sees **both** the decoder state  $s_u$  **and** the freshly-attended source context  $c_u$

Query the encoder states, softmax the scores into weights, take the weighted sum:



thicker edge = larger  $\alpha$ ; here step  $u$  attends mostly to  $h_3$ , so  $c_u \approx h_3$

# Attention is a differentiable lookup

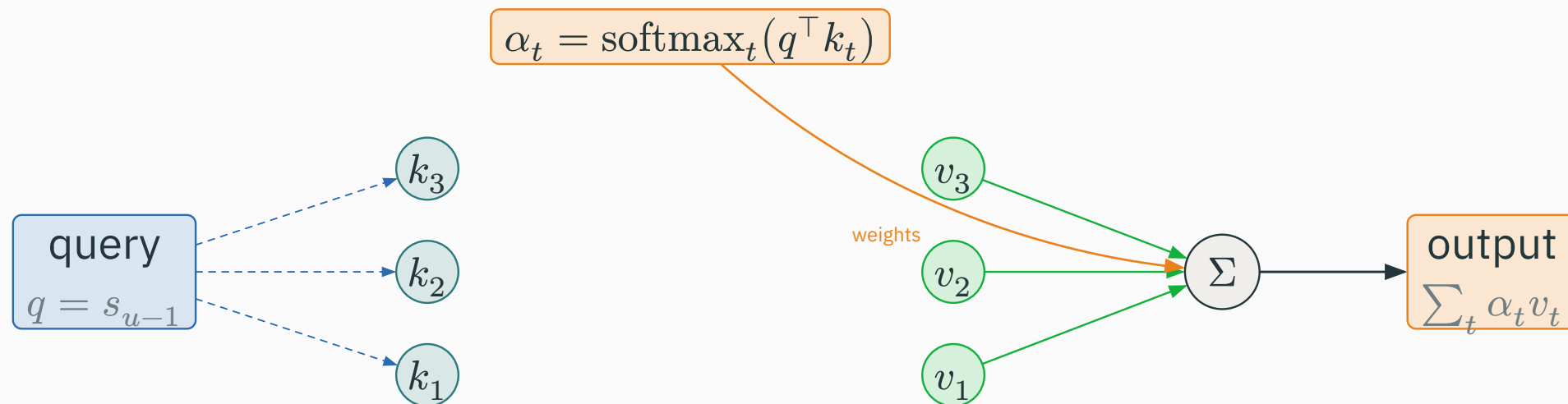
Rename the pieces and attention becomes a **soft dictionary lookup** — the language of Transformers:

**query**  $q = s_{u-1}$  — what the decoder is looking for.

**keys**  $k_t = h_t$ , **values**  $v_t = h_t$  — what the source offers.

$$\alpha_t = \text{softmax}_t(q^\top k_t), \quad \text{output} = \sum_{t=1}^T \alpha_t v_t$$

# The lookup, drawn



a **hard** lookup returns one value; attention returns a **soft, weighted** blend — and is differentiable, so it **learns**

# **A worked attention step**

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Scalar encoder states  $h = [1, 2, 4]$  and a query  $q = 2$ . Dot-product scores:

$$e_t = q \cdot h_t : \quad e = [2, 4, 8]$$

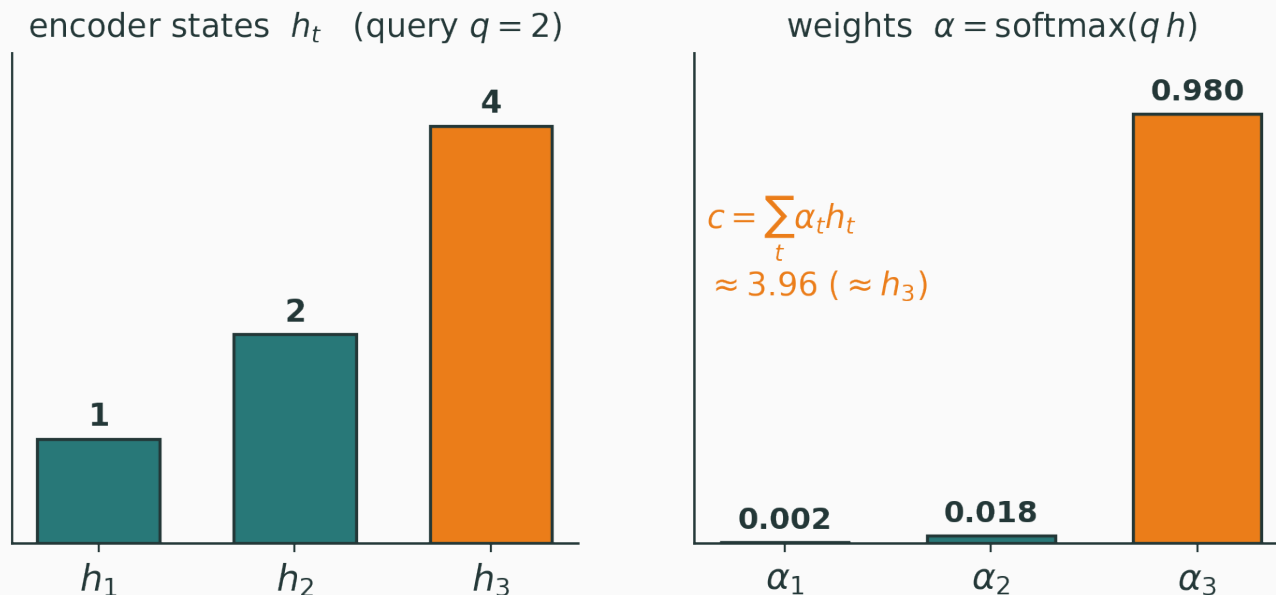
Softmax the scores into weights:

$$\alpha = \text{softmax}([2, 4, 8]) \approx [0.0024, 0.0179, 0.9796]$$

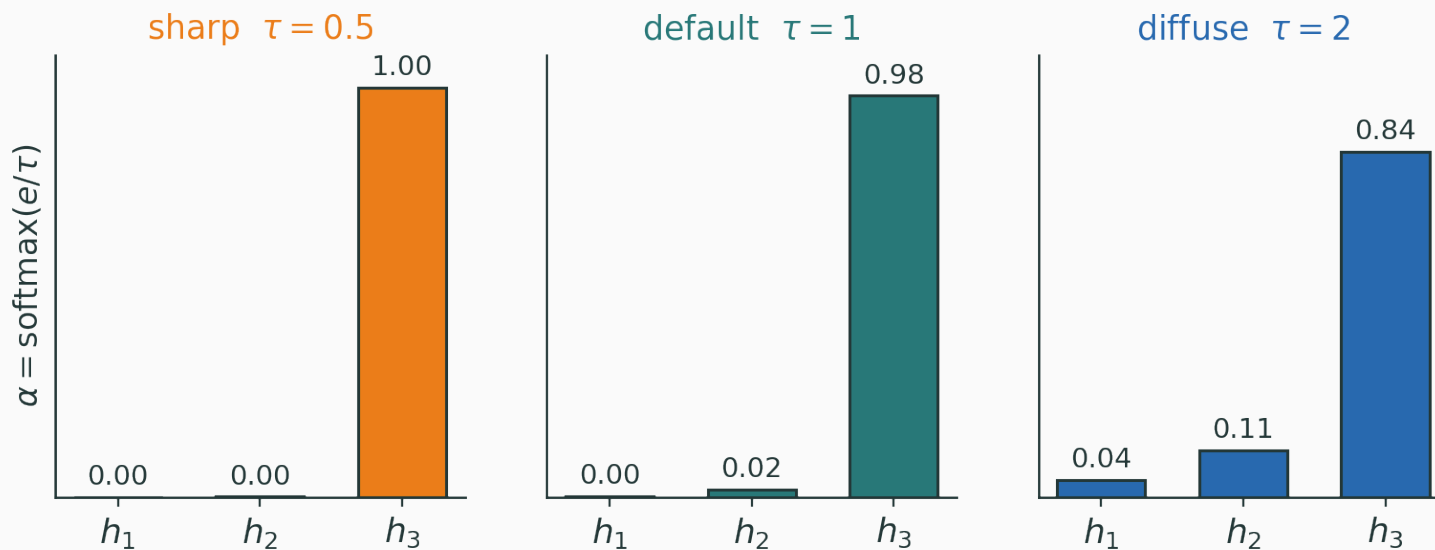
the score gap of 8 vs 4 is **exponentiated** — step attends almost entirely to  $h_3$

Weighted sum of the states — the context lands near the most-attended one:

$$c = \sum_t \alpha_t h_t \approx 0.0024(1) + 0.0179(2) + 0.9796(4) \approx 3.96$$



Softmax with temperature  $\tau$ :  $\alpha = \text{softmax}(e/\tau)$  — small  $\tau$  sharpens, large  $\tau$  spreads:



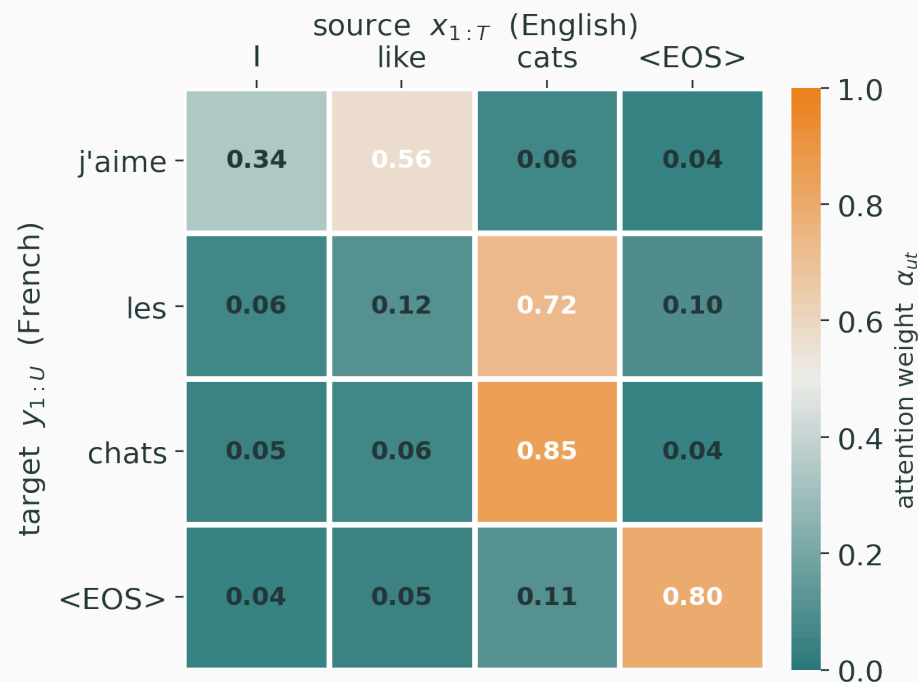
Transformers use **scaled** dot-product  $q^\top k / \sqrt{d}$  — the  $\sqrt{d}$  keeps scores from saturating the softmax

# **Alignment and interpretability**

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# The attention matrix

Stack the weights over all steps:  $A \in \mathbb{R}^{U \times T}$ , with  $A_{ut} = \alpha_{ut}$ . Each row is one output step's distribution over the source:



The bright cells trace a soft **word alignment** — learned, never supervised:

| <b>target token</b> | <b>attends to source</b>     |
|---------------------|------------------------------|
| j'aime              | like (and a little I)        |
| les                 | cats (the plural determiner) |
| chats               | cats                         |
| <EOS>               | <EOS>                        |

attention discovers **which source word each target word corresponds to** — for free

**Caution.** Attention weights are a useful **diagnostic** of what the model looked at — but they are **not always a faithful explanation**. Different weightings can give the same output, and high weight  $\neq$  causal importance (Jain & Wallace 2019).

## INTERACTIVE

↗ [nipunbatra.github.io/interactive-articles/attention](https://nipunbatra.github.io/interactive-articles/attention)

Hover the alignment heatmap for “I like cats” → “j’aime les chats”: watch each target row light up the source words it attends to, and edit the sentence to see the alignment shift.

# **The full seq2seq + attention model**

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The encoder is unchanged — but now we **keep all** the states, not just the last one:

$$h_t = f_{\text{enc}}(h_{t-1}, e(x_t)), \quad H = (h_1, \dots, h_T)$$

$H$  is the memory the decoder attends into — the fixed  $c = h_T$  is gone

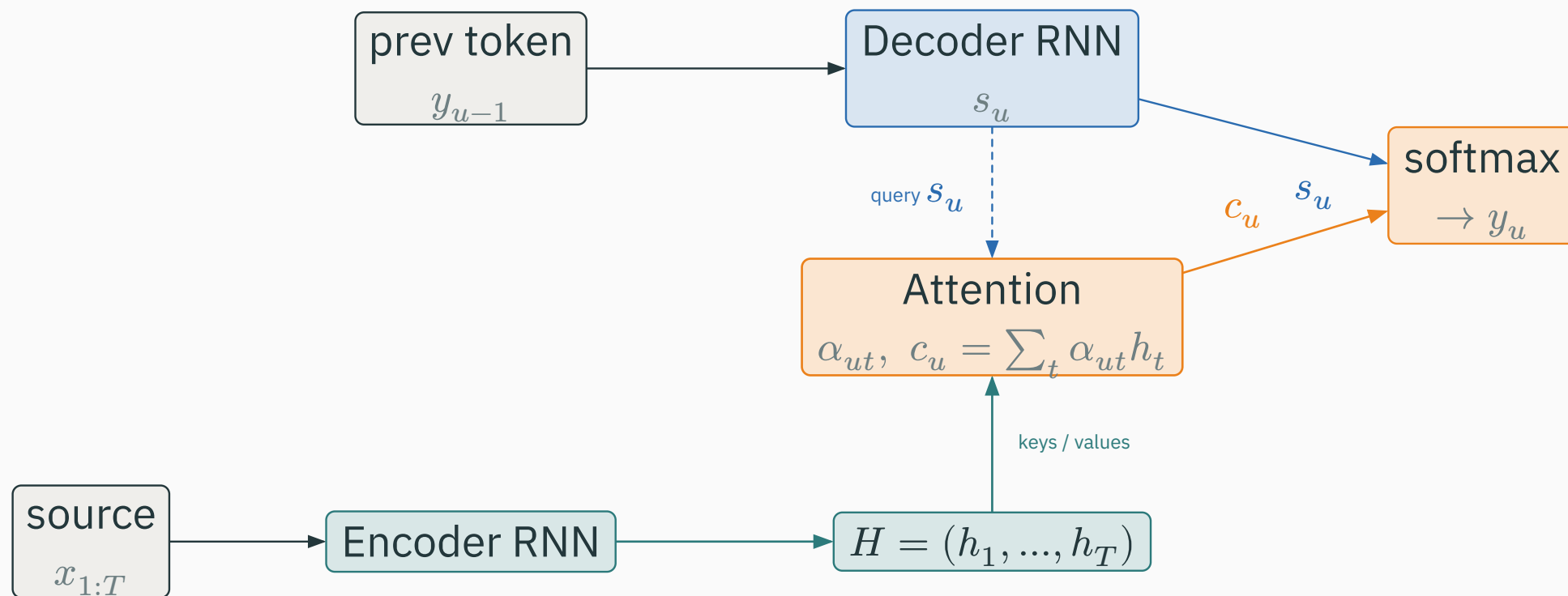
Each decoder step: update the state, attend over  $H$ , predict from state + context:

$$s_u = f_{\text{dec}}(s_{u-1}, e(y_{u-1}), c_{u-1})$$

$$e_{ut} = \text{score}(s_u, h_t), \quad \alpha_{ut} = \text{softmax}_t(e_{ut}), \quad c_u = \sum_{t=1}^T \alpha_{ut} h_t$$

$$p(y_u \mid \cdot) = \text{softmax}(W_o [s_u; c_u] + b_o)$$

Encoder produces  $H$ ; every decoder step queries it for a fresh context  $c_u$ :



the attention block runs **once per output step** — a new  $c_u$  each time, chosen by the query  $s_u$

Same masked cross-entropy as before — but now the gradient reaches **further**:

$$\mathcal{L} = - \sum_{u=1}^U \log p(y_u \mid y_{<u}, x_{1:T})$$

Backprop flows: decoder readout  $\rightarrow$  attention weights  $\alpha_{ut}$   $\rightarrow$  encoder states  $h_t \rightarrow$  the encoder RNN. Attention is **fully differentiable**, so the alignment is **learned** end-to-end from the translation loss alone.

# Implementation

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Sequences differ in length, so **pad** to a common  $T$  and track a **source mask**:

$$m_t = \begin{cases} 1 & x_t \neq \text{<PAD>} \\ 0 & x_t = \text{<PAD>} \end{cases}$$

the mask marks which source positions are real — attention must not attend to padding

Padding positions must get **zero** weight. Set their scores to  $-\infty$  **before** the softmax:

$$e_{ut} \leftarrow \begin{cases} e_{ut} & m_t = 1 \\ -\infty & m_t = 0 \end{cases} \Rightarrow \alpha_{ut} = 0 \text{ on padding}$$

$\exp(-\infty) = 0$ , so masked positions drop out of the softmax cleanly and the weights still sum to 1 over the **real** tokens. (The same trick gives the **causal** mask in Transformers.)

Train step: feed the **true** previous target each step, accumulate masked cross-entropy:

```
H, enc_state = encoder(src, src_mask)    # keep ALL states H for attention
dec_in = BOS.expand(batch)               # first decoder input
state = init_from(enc_state); loss = 0
for u in range(U):                       # teacher forcing along the target
    logits, state, alpha = decoder(dec_in, state, H, src_mask)
    loss += cross_entropy(logits, tgt[:, u], ignore_index=PAD)
    dec_in = tgt[:, u]                   # feed the TRUE token, not argmax
loss.backward()
```

Inference: feed the model's **own** output back; or keep  $B$  hypotheses:

```
# greedy
dec_in = BOS; out = []
for _ in range(max_len):
    logits, state, _ = decoder(
        dec_in, state, H, src_mask)
    dec_in = logits.argmax(-1)
    if dec_in == EOS: break
    out.append(dec_in)
```

```
# beam search (B hypotheses)
beams = [Hyp(tokens=[BOS], state=s0,
              score=0.0, done=False)]
# each step: expand every beam,
#   score = cum log-prob,
#   keep top-B, stop when all done
```

# **Limitations and summary**

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# What attention fixed — and did not

**Fixed** — the bottleneck.

before:  $x \rightarrow c = h_T \rightarrow y$ .

after:  $x \rightarrow H$ , and  $c_u = \sum_t \alpha_{ut} h_t$  **every step**.

**Not fixed** — recurrence. The decoder is **still** an RNN:  $s_u = f(s_{u-1}, y_{u-1}, c_{u-1})$  — target generation stays **sequential**.

attention removed **one** bottleneck; the **sequential** encoder and decoder remain

- the encoder is  $O(T)$  **sequential** — cannot parallelize over source time;
- the decoder is  $O(U)$  **sequential** — each  $s_u$  needs  $s_{u-1}$ ;
- long-range dependencies still travel through many recurrent steps.

**if attention can access any position directly — do we even need recurrence?**

# What changed, across three lectures

| Model               | Conditional                             | Context used           |
|---------------------|-----------------------------------------|------------------------|
| RNN LM              | $p(y_t \mid y_{<t})$                    | own past only          |
| seq2seq             | $p(y_t \mid y_{<t}, x)$                 | one fixed $c = h_T$    |
| seq2seq + attention | $p(y_t \mid y_{<t}, \sum_t \alpha h_t)$ | per-step context $c_u$ |

each row adds access: to the source, then to **every source position, dynamically**

| Idea            | Takeaway                                                                |
|-----------------|-------------------------------------------------------------------------|
| encoder–decoder | encode source $\rightarrow c$ ; decode target autoregressively from $c$ |
| teacher forcing | feed true $y_{u-1}$ in training — stable, but <b>exposure bias</b>      |
| decoding        | greedy / beam search; length-normalize the score                        |
| bottleneck      | one fixed $c = h_T$ loses long, structured sources                      |
| attention       | $c_u = \sum_t \alpha_{ut} h_t$ — per-step, source-dependent context     |
| scoring         | dot / general / additive; softmax; query–key–value lookup               |

# Final mental model — one idea

The **encoder** represents the source; the **decoder** generates the target.

**Teacher forcing** trains it; **attention** lets each step read the source **dynamically**.

attention = a differentiable, weighted lookup over all source states

Attention removed the bottleneck — but recurrence remains.

Next: **Transformers** — remove recurrence entirely and make **attention the main computation** (queries, keys, values, self-attention, causal masks, positional encodings).